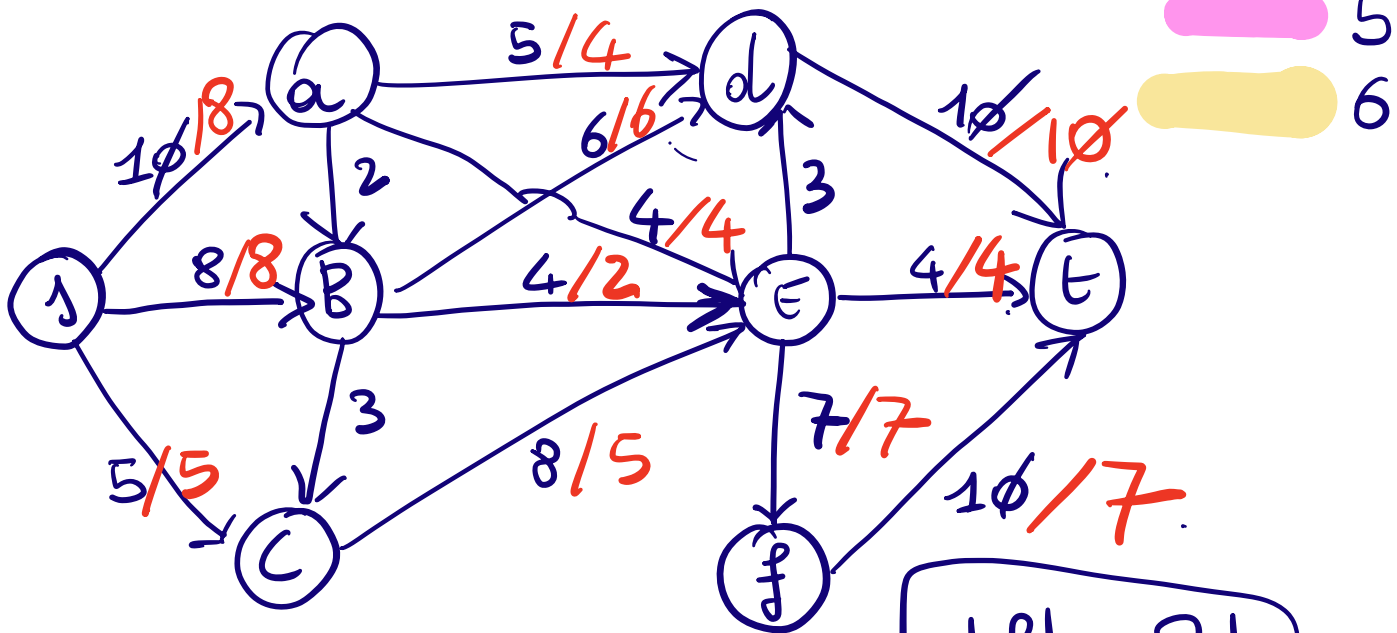
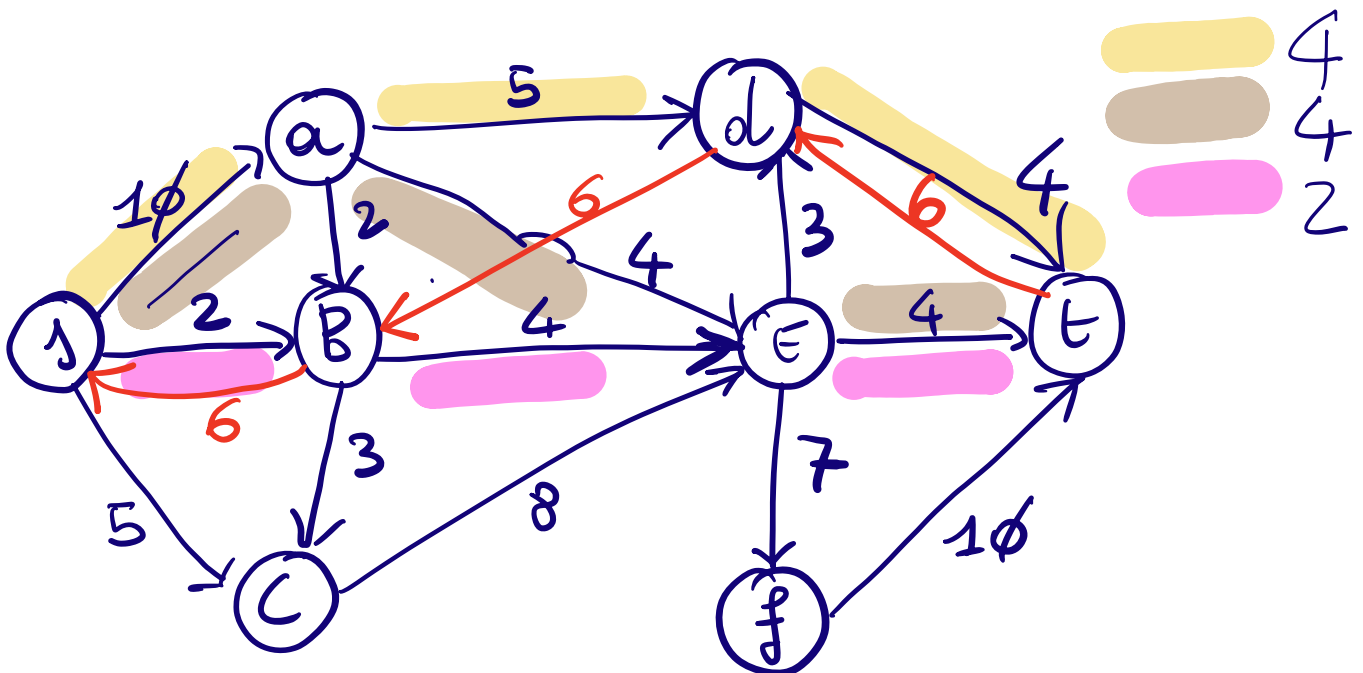


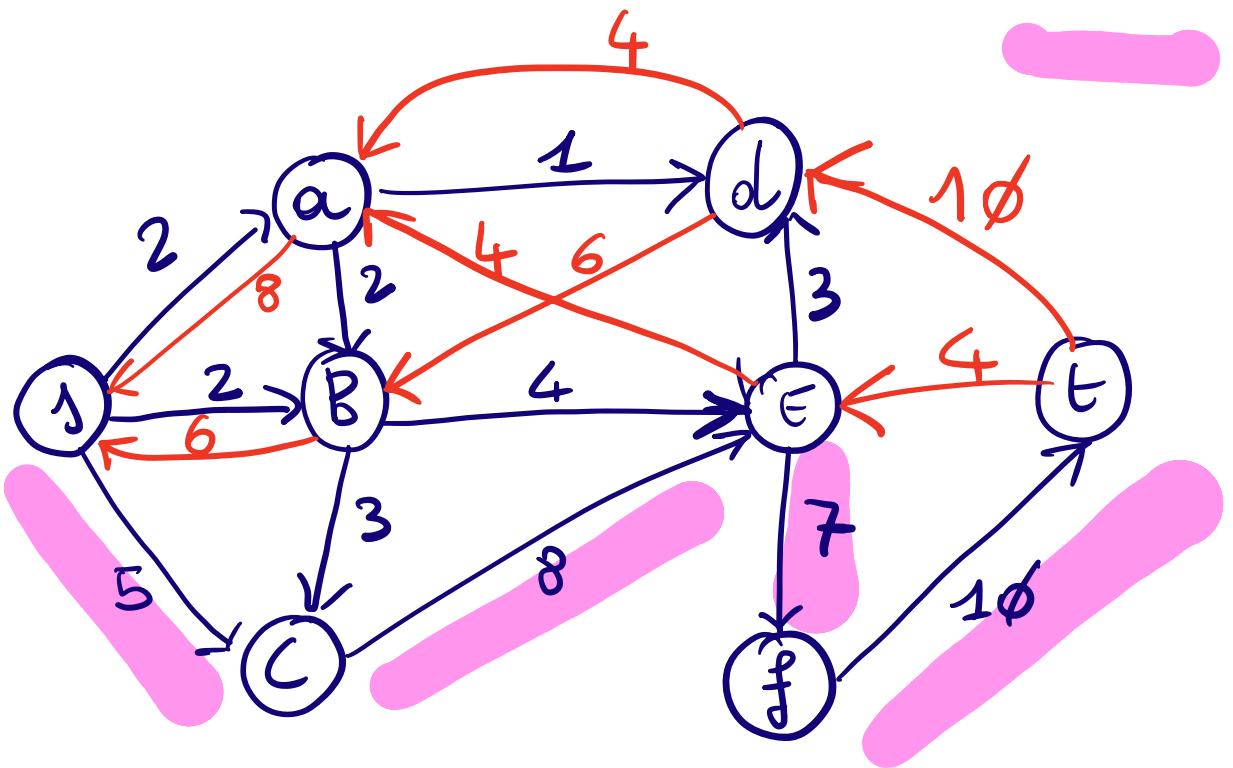
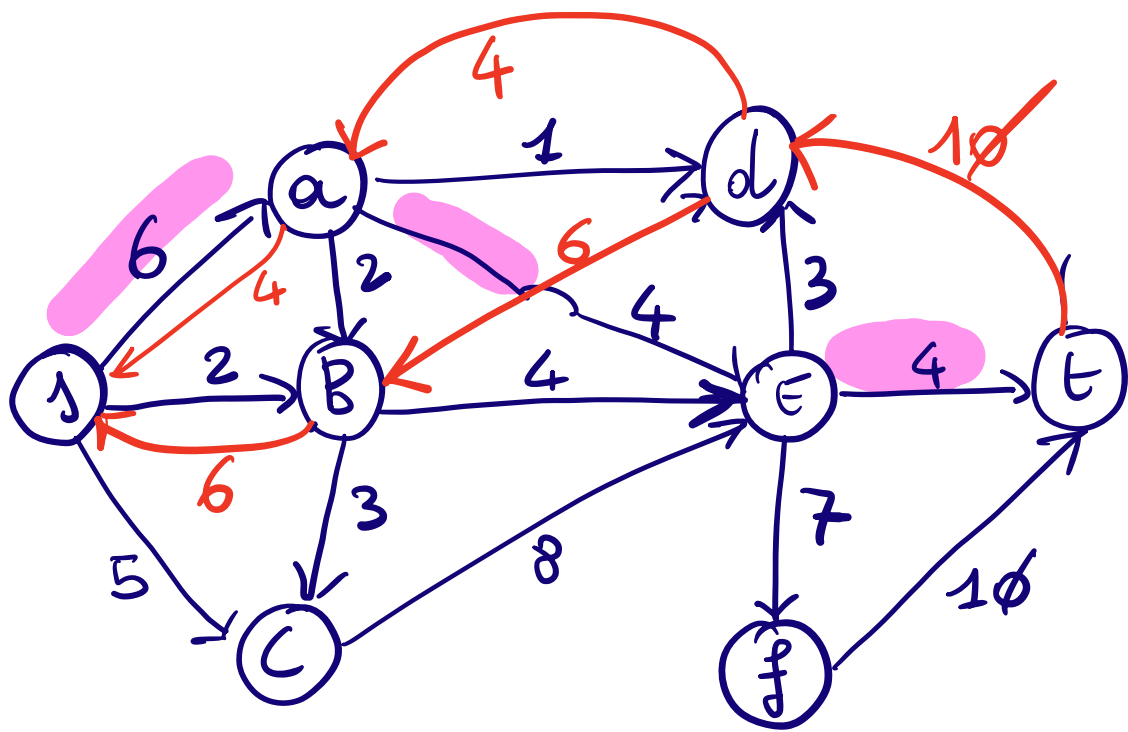


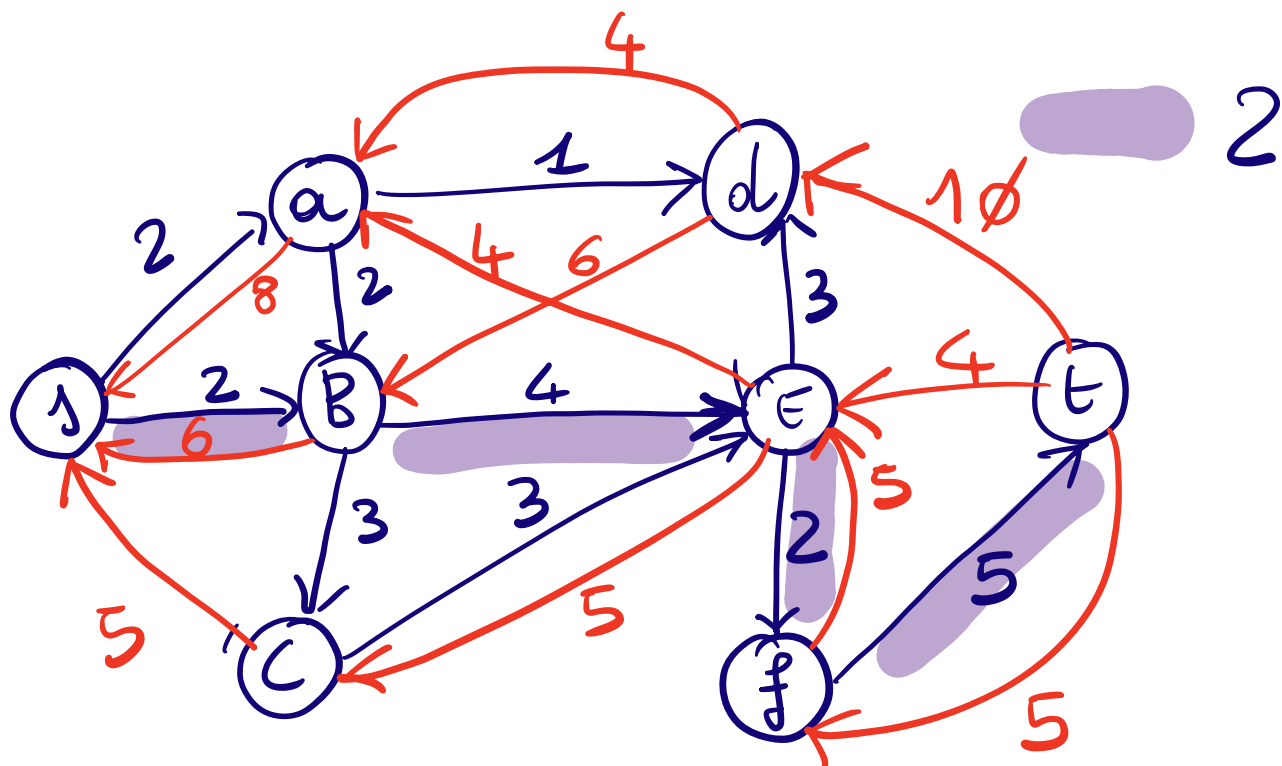
$$|f| = 21$$

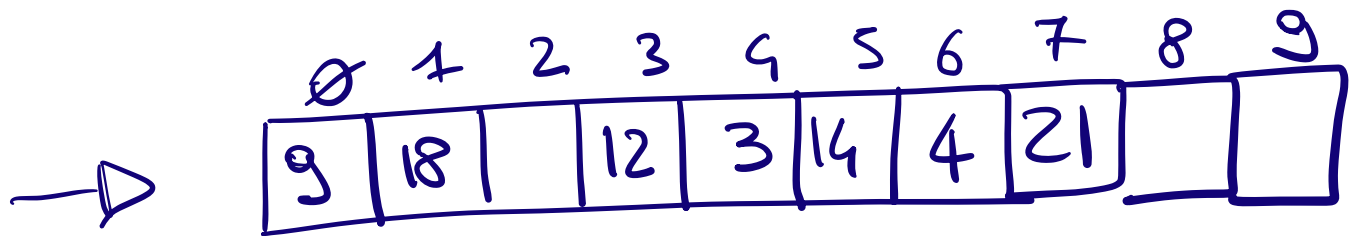
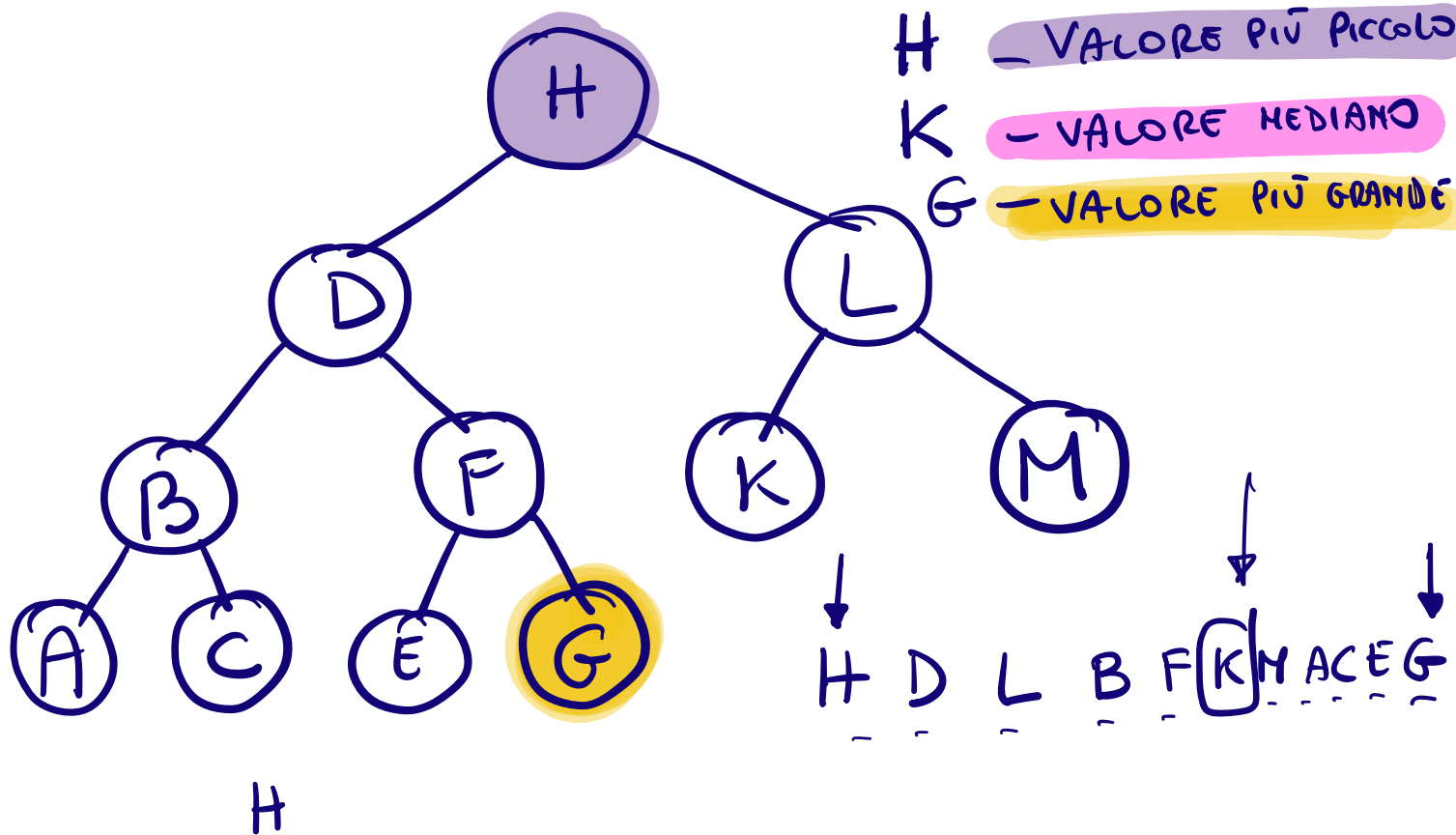
$s - b - e - t$

$\times s - b - e - f - t \quad \leftarrow$









$$h(x) = x \bmod m$$

X(a) 9, 14, 4, 18, 12, 3, 21 ←

X(b) 12, 3, 14, 18, 4, 9, 21

X(c) 12, 14, 3, 9, 4, 18, 21

X(d) 9, 12, 14, 3, 4, 21, 18

X(e) 12, 9, 18, 3, 14, 21, 4

A

$$T(n) = 4T(n/2) + \boxed{\frac{n}{\log n}}$$

$$n^{\log_b a} = n^{\log_2 4} = \boxed{n^2}$$

~~CASO 2~~

CASO 1

$$\Theta(n^{\log_b a})$$

CASO 3

$$\Theta(n^2)$$

$$\frac{n}{\log n}$$

$$n^2$$

25/7/2023 pseudocode(D8)

$$T(n) = 8T\left(\frac{n}{3}\right) + \underbrace{2n^2 \log n}$$

$$n^{\log_a b} = n^{\log_3 8} \quad 1 < \log_3 8 < 2$$

$$\underbrace{2n^2} \cdot \log n \sim n^x \quad 1 < x < 2$$

$$\text{CASO 3} \quad 2n^2 \cdot \log n = \Omega\left(n^{\log_3 8 + \epsilon}\right)$$

$$\underbrace{a f\left(\frac{n}{b}\right) \leq c \cdot f(n)}_{b < c < 1}$$

$$8 \cdot \cancel{2} \cdot \left(\frac{n}{3}\right)^{\cancel{2}} \cdot \log\left(\frac{n}{3}\right) \leq c \cdot \cancel{2} n^{\cancel{2}} \cdot \log n$$

$$\frac{8}{9} \cdot \log\left(\frac{n}{3}\right) \leq c \cdot \log(n)$$

$$\Theta(2 \cdot n^2 \cdot \log n)$$

$$c = \frac{8}{9}$$

$$\Theta(n^2 \cdot \log n)$$